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Top Skills

Advanced Automation
Cloud Computing
Data Integration

Certifications

Pre Security
[PCEP-30-01] PCEP – Certified
Entry-Level Python Programmer
Offensive Pentesting
Google IT Automation with Python
Advent of Cyber 2025 Competition

Mangesh Bhattacharya

Cybersecurity#Practitioner#|#MIT-S-AI#Graduate#Student#|#Azure
& AWS Security#Practitioner#| AI#Automation#| Web Penetration
Tester#|#Machine#Learning#for#Threat#Detection
Canada

Summary

Cybersecurity Engineer and MIT-S-AI graduate student building the next generation of AI-driven, cloud-native security systems. I design detection and response architectures that combine machine learning, automation, and deep cloud security expertise across AWS and Azure. Currently working as a freelance Security Engineer on Upwork, where I strengthen cloud security posture, support SOC#2#readiness, and build scalable, secure infrastructure for engineering teams. My background blends hands-on SOC,#DevSecOps,#and#Threat#Intelligence experience with applied research in AI for Cybersecurity, focusing on autonomous defense and real-time anomaly detection. Selected Work Highlights: Engineered an ML-based intrusion detection system achieving#90%+#accuracy in anomaly detection. Built an AI-powered malware classifier that improved detection speed by#35%. Designed automated threat-intelligence pipelines that reduced manual review time by#40%. Implemented AWS#and#Azure#security monitoring to detect and contain privilege-escalation attacks. Core Toolkit: Python#|#C++ | Bash | AWS#|#Azure#|#SIEM/SOAR/ NIST#|#MITRE#ATT&CK#|#IAM#|#Zero#Trust#Architecture#|#ML#f Cisco#Ethical#Hacking#|#CCNA#|#Google#IT#Automation#with#Python# by curiosity and innovation, I'm passionate about advancing AI-security integration, autonomous defense, and cyber resilience. Always open to connecting with professionals working at the intersection of AI,#Cybersecurity,#and#Cloud#Engineering to push detection, automation, and defense forward.

Experience

Upwork
1 year

Bug Bounty Hunter
August 2025 - Present (1 year)

Identify and analyze security vulnerabilities in web and mobile applications using industry-standard penetration testing tools and methodologies.

Conduct in-depth vulnerability assessments and exploit validation using Burp#Suite,#OWASP#ZAP,#Wireshark, and custom scripts for dynamic analysis.

Deliver comprehensive technical documentation and detailed vulnerability reports with clear replication steps and actionable remediation guidance.

Collaborate with development teams to understand application architecture, business logic, and prioritize fixes based on impact and exploitability.

Perform manual and automated testing aligned with OWASP#Top#10, CWE, and MITRE#ATT&CK frameworks to ensure robust application security.

Participate in Capture#the#Flag#(CTF) competitions and bug bounty programs to continuously refine offensive security skills and stay current with emerging threats.

Apply ethical hacking, exploit development, and reconnaissance techniques to uncover misconfigurations, injection flaws, and privilege escalation paths.

Maintain strong focus on responsible disclosure, secure coding practices, and continuous learning through active engagement with global security communities.

Cybersecurity Engineer (Freelance)

August 2025 - Present (1 year)

Toronto, ON

Apply heuristic analysis and machine learning to architect production-grade User and Entity Behavior Analytics (UEBA) for real-time anomaly detection;

Design and execute high-fidelity adversarial emulations leveraging MITRE ATT&CK frameworks and custom payload obfuscation to bypass EDR/XDR solutions;

Engineer end-to-end SOAR (Security Orchestration, Automation, and Response) workflows using Python and TensorFlow to accelerate Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR);

Conduct deep-dive attack surface management using graph theory and network topology visualization to identify hidden lateral movement vectors and privilege escalation paths;

Integrate cybersecurity observability with AI-driven threat modeling, bridging the gap between Black Hat exploitation techniques and Green Hat (ethical) remediation orchestration;

Optimize SIEM (Security Information and Event Management) ingestion and log correlation to ensure technical GDPR compliance, focusing on data pseudonymization and encryption at rest/transit;

Collaborate with enterprise clients to translate Identity and Access Management (IAM) requirements into least-privilege algorithmic solutions, utilizing Zero Trust principles to harden perimeter-less environments.

Cyber Security Engineer

August 2025 - Present (1 year)

Conduct web and mobile penetration tests using Burp Suite, Nmap, OWASP ZAP, FFUF and Nuclei. Develop security automation workflows with Python, SOAR platforms, Snyc and Trivy. Develop SIEM/UEBA solutions and automate incident response workflows. Perform MITRE ATT&CK emulation and deliver remediation reports.

DevSecOps Engineer (Freelance)

August 2025 - January 2026 (6 months)

Toronto, ON

Built automated vulnerability-scanning pipelines that accelerated detection workflows and expanded coverage across client applications and cloud environments.

Delivered clear, actionable remediation reports with reproducible steps, enabling clients to resolve issues efficiently and strengthen their security posture.

Partnered with development teams to prioritize fixes, streamline remediation cycles, and reduce average time-to-patch across CI/CD pipelines.

Performed web and mobile application security assessments, reducing exploitable vulnerabilities and improving client confidence in deployed systems.

Provided ongoing security guidance and best-practice training, empowering clients to maintain secure development and operational processes.

Enhanced client environments by integrating advanced threat-detection and monitoring tools, resulting in fewer security incidents and improved visibility.

Executed comprehensive audits of existing security controls, identifying critical gaps and implementing targeted improvements to reinforce system integrity.

Designed custom security training programs tailored to client needs, increasing awareness, preparedness, and proactive risk management.

TryHackMe

CTF Player

May 2021 - May 2026 (5 years 1 month)

Ontario, Canada

Passionate and results-driven cybersecurity professional specializing in the hands-on application of security principles through Capture-the-Flag

(CTF) challenges and real-world simulations. Experienced in identifying vulnerabilities, exploiting weaknesses, and strengthening system defenses across diverse environments.

Core Competencies:

Offensive#Security#&#Penetration#Testing

Vulnerability#Assessment#&#Exploitation

Linux#and#Windows#System#Internals

Network#Reconnaissance#&#Traffic#Analysis

Threat#Hunting#&#Incident#Response

Ethical#Hacking#&#Red#Team#Operations

Actively pursuing roles such as

Penetration#Tester,#Red#Teamer,#Blue#Teamer,#or#SOC#Analyst to apply analytical expertise and deliver secure outcomes for organizations. Committed to continuous learning, CTF participation, and collaboration with cybersecurity professionals to advance modern defense strategies.

Ameya Data Solutions

1 year 1 month

Cybersecurity Observability & AI Engineer

December 2024 - May 2025 (6 months)

Mississauga, ON

Designed and deployed AI-powered threat detection and observability systems using machine learning and anomaly modeling to enhance SOC visibility and reduce false positives.

Built real-time monitoring pipelines integrating SIEM,#IDS,#IPS,#EDR, and cloud telemetry for unified visibility, proactive threat hunting, and automated incident response.

Developed custom dashboards and alerting frameworks

(OpenSearch,#Kibana,#Python) for real-time threat visualization and executive-level reporting.

Engineered endpoint detection and response (EDR) automation and AI-driven threat-response workflows to improve detection accuracy and streamline operations.

Strengthened AWS#cloud security through IAM#policy design,

VPC#segmentation, Security#Groups, and Zero-Trust architecture aligned with NIST,#CIS,#and#GDPR standards.

Automated vulnerability management, patching, and compliance checks using Ansible#and#Python, improving efficiency and consistency across environments.

Integrated multi-source logs into OpenSearch#Dashboards for forensic analysis, anomaly detection, and accelerated incident response.

Implemented Docker-based secure deployments to reduce configuration drift and enhance container isolation.

Conducted penetration testing, cloud security assessments, and vulnerability scanning to identify and remediate security gaps.

Collaborated with SOC,#DevSecOps,#and#Cloud#Engineering teams to translate security requirements into scalable analytical and AI-driven detection solutions.

Authored technical documentation, training materials, and cybersecurity awareness content to strengthen organizational security culture.

DevSecOps Engineer

May 2024 - November 2024 (7 months)

Mississauga, ON

Developed AI-enhanced EDR tools using Python,#Snyk,#and#Trivy to improve vulnerability detection and automate CVE-level threat response.

Designed and enforced AWS#security groups,#IAM#policies,#and#VPC#architectures, strengthening access control and ensuring compliance with NIST#and#GDPR standards.

Deployed Fluent#Bit and integrated system logs into OpenSearch#Dashboards, enhancing real-time visibility and accelerating forensic investigations.

Automated security configurations, patching, and compliance checks using Ansible#and#AI-driven workflows, ensuring consistent, secure deployments across cloud environments.

Implemented Docker-based containerization to secure application delivery pipelines, reduce configuration drift, and improve deployment efficiency.

Created custom threat-monitoring dashboards for automated incident response, optimizing detection and resolution times.

Conducted cloud security assessments and remediation to support a Zero-Trust#infrastructure and strengthen overall security posture.

Authored technical documentation and training materials to promote cybersecurity awareness and best practices across engineering teams.

Ameya Data Solutions

Data Analyst Intern

August 2023 - January 2024 (6 months)

Mississauga, ON

Designed and implemented AWS#security groups to monitor and control application traffic, improving visibility and compliance across cloud environments.

Developed Fluentd log-management packages and deployed OpenSearch Dashboards, integrating system and access logs into Kibana (ELK Stack) for enhanced analytics and operational insight. Managed AWS cloud infrastructure (EC2, S3, VPC, IAM) to ensure resilient, scalable, and secure deployments aligned with organizational standards. Automated security configuration and deployment workflows using Ansible playbooks and Docker, ensuring consistency and reducing manual overhead.

Utilized Python for data cleaning, manipulation, and visualization to support analytical reporting and decision-making.

Built custom dashboards for real-time threat monitoring and incident response, improving detection and response efficiency.

Conducted vulnerability assessments and system hardening across Windows, Linux, and macOS environments to strengthen endpoint security.

Developed intrusion detection and prevention systems, firewalls, and network-security measures to safeguard data integrity.

Authored training manuals and documentation for technical and non-technical audiences, promoting cybersecurity awareness and best practices.

STAPLES Canada

Technology Associate

November 2016 - August 2019 (2 years 10 months)

Toronto, Canada Area

Provided technical support and troubleshooting for hardware and software issues, demonstrating effective communication and problem-solving skills.

Assisted with basic IT security practices and configuration, reinforcing foundational security knowledge in a customer-facing context.

Explained technical concepts to non-technical users in clear, professional language and collaborated with colleagues to maintain high service standards.

Staples

Customer Service Representative

January 2016 - November 2016 (11 months)

Toronto, Canada Area

Education

Ontario Tech University

Master of Information Technology Security, Artificial Intelligence · (September 2025 - April 2027)

Seneca Polytechnic

Honours Bachelor of Information Technology, Cybersecurity · (May 2021 - August 2025)

Humber Polytechnic

Engineering Diploma, Computer Systems Networking and Telecommunications · (September 2018 - May 2021)

Seneca College of Applied Arts and Technology

Ontario College Diploma, Aviation Safety and Operations · (2014 - 2018)

Xyna International High School

High School · (2011 - 2014)